Prototype refinement

Iterative refinement

Support prototypes

Soft assignments

Weighted soft prototypes

Refined prototypes

Blend

Weighted mean

Unlabeled features

Retain support anchors

Sharpening

Class prompts P

Fixed signed gates

Node-wise correction

Post-training: neighbors and propagation

Embedding similarity

Neighbor selection

Representation neighbor graph

a

b

c

Retrieve by similarity

Exclude self

Similarity weights

Prototype classifier

Original probabilities F_0

Keep fixed

Fixed W

Shared frozen graph encoder

LeakyReLU

Input graph

Layer 1

Hidden-feature correction

Layer 2

Adapted representations

Current probabilities

Updated probabilities

Neighbor propagation

Iterative update

Original F_0

10%

Weighted neighbor probabilities

90%

Updated class probabilities

Clamp support labels

Predicted classes

Prompt learning: direction and norm adaptation

Support loss

Shared gradient G

Rescaled direction $\tilde{D}$

Same ray; only the length changes

Adam reference A

Use its norm, not its direction

$$\|\tilde{D}\|_F = \|A\|_F$$

Match the whole-matrix norm

Shampoo direction D

Common origin

Dashed-circle radius = Adam norm

$$P \leftarrow P - \eta\tilde{D}$$

Parameters move in the negative direction

Frozen backbone

Support node

Only P is updated